



Mmfpoga12

Start Test Applications - Manual

REV 1.0

1. Connect PLDcable II programmer to your computer by LPT connector and to JTAG connector placed on Mmfpga1x module.
2. Connect supply adaptor to your Evaluation Board.
3. Power on Evaluation Board.
4. Start Xilinx ISE or Xilinx WebPack environment (it's free and available from side www.xilinx.com). After that our screen will be looking as below in Figure 1:

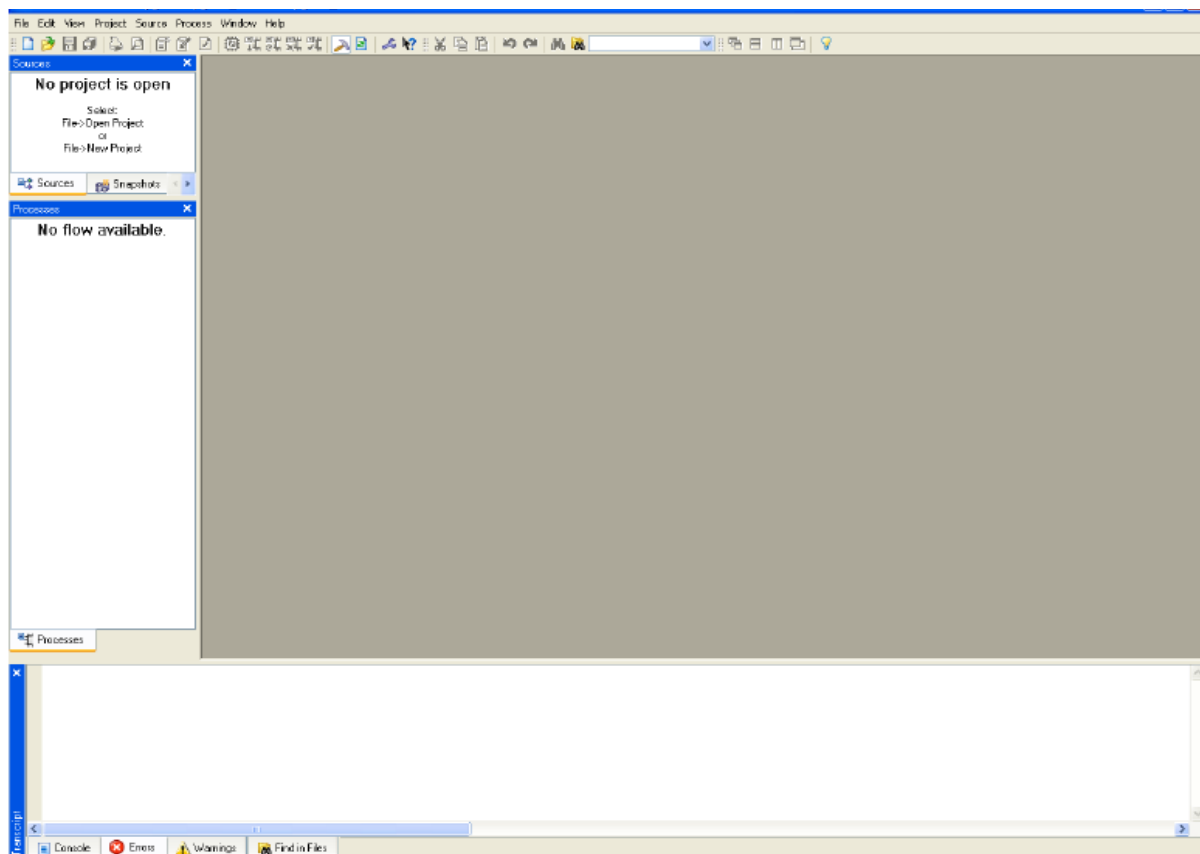
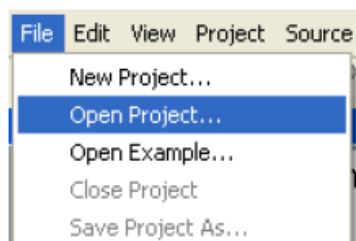


Fig. 1

5. Open one of test projects available form CD (**Path**). For example selecting project Test_1 choose File -> Open Project...



Is showing Open Project window (Figure 2), in which we have to select our project *Test_1.isc* and accept.

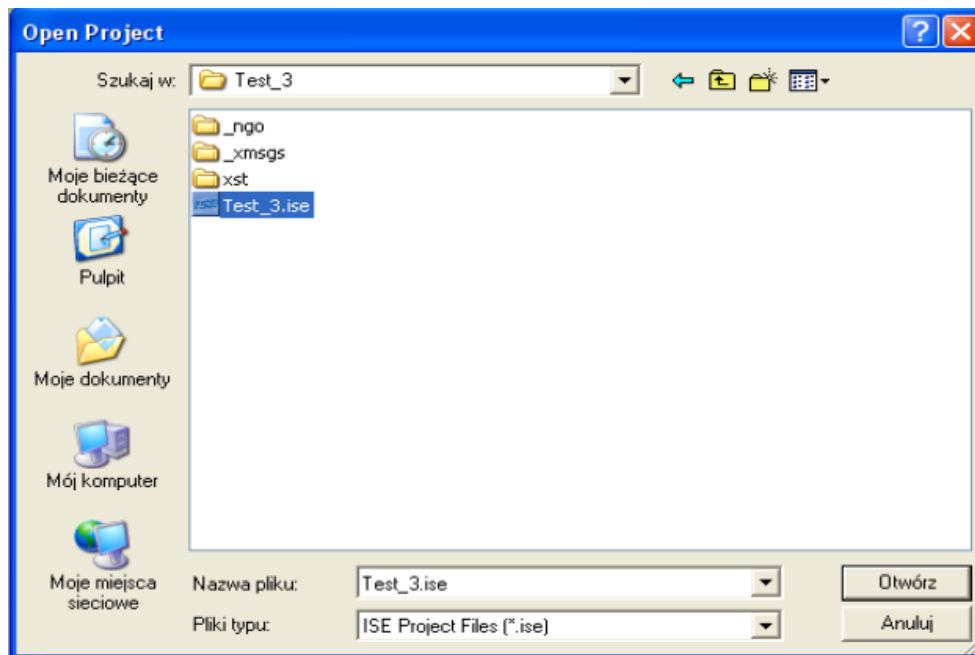


Fig. 2

6. Our project is opened, and window is looking as in Figure 3 below:

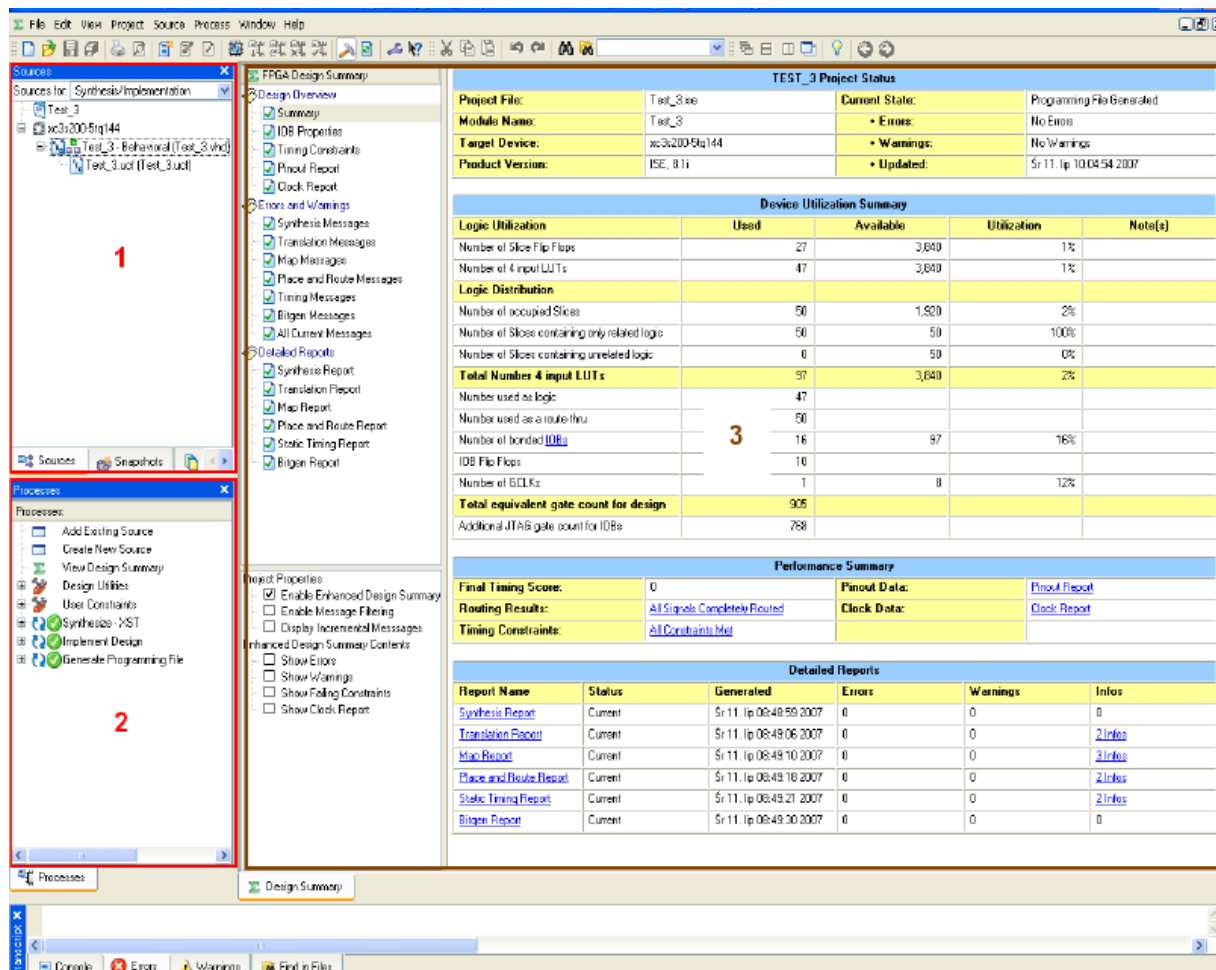


Fig. 3

In window number 1 we can see project three and files included to our project (*Test_3.vhd and Test_3.ucf*):

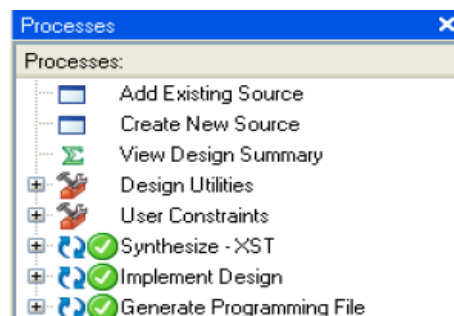
- *.vhd – include VHDL code
- *.ucf – include Spartan 3 pins allocation to our ports declared in entity
(**must always be included to project !!!**)

Window number 2 include available process which we can make.

Window number 3 is a major window in which we can for example editing our files.

On the bottom we have console window, where we can observe synthesize and implementation progress or some errors and warnings.

7. Clicking twice in window 1 on file *Test_3.vhd* and in window 3 should appear VHDL code of our project. In window 2 we can see processes available for our project file:

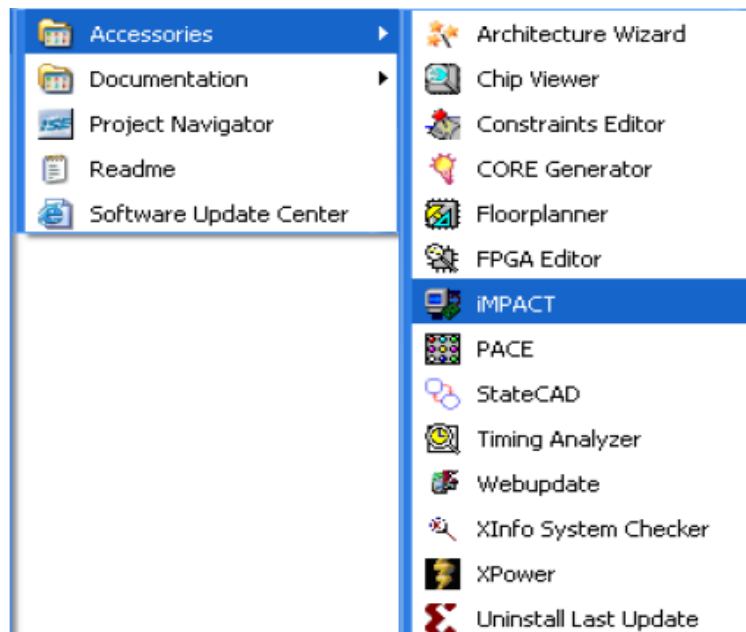


It's synthesize, implementation and programming file generating which later will be use to programme our chip. Each of that process can be started alone by dable click on it and observe progress in consloe window. When we want do all processes together, simply clicking twice on last process and two previous will be done automatically.

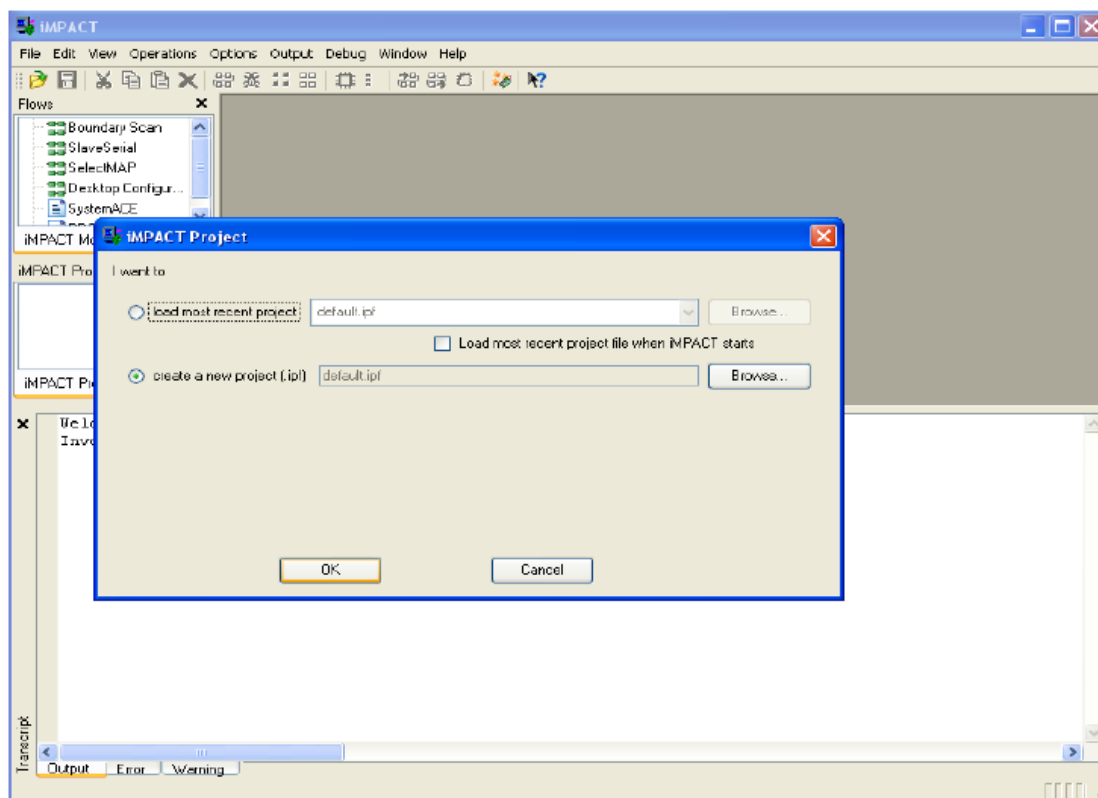
After every third processes will be done, should be appear green chack (everything is ok) or yellow warnings (can be observe in console window) by the left of each process.

In major folder of our project should appear file *Test_3.bit* , which is a programming file using to programme our chip.

8. To programme our chip, please start application iMPACT by choose from Start menu Programs, Xilinx ISE X.Xi and *Accessories* -> *iMPACT*.



Will be appear window:



Select '*create a new project*' and click OK. When next window appear don't change nothing and click Finish. Application should detect our chip. On the end please close two next windows by Cancel. It should be look as in the Figure 4 below.

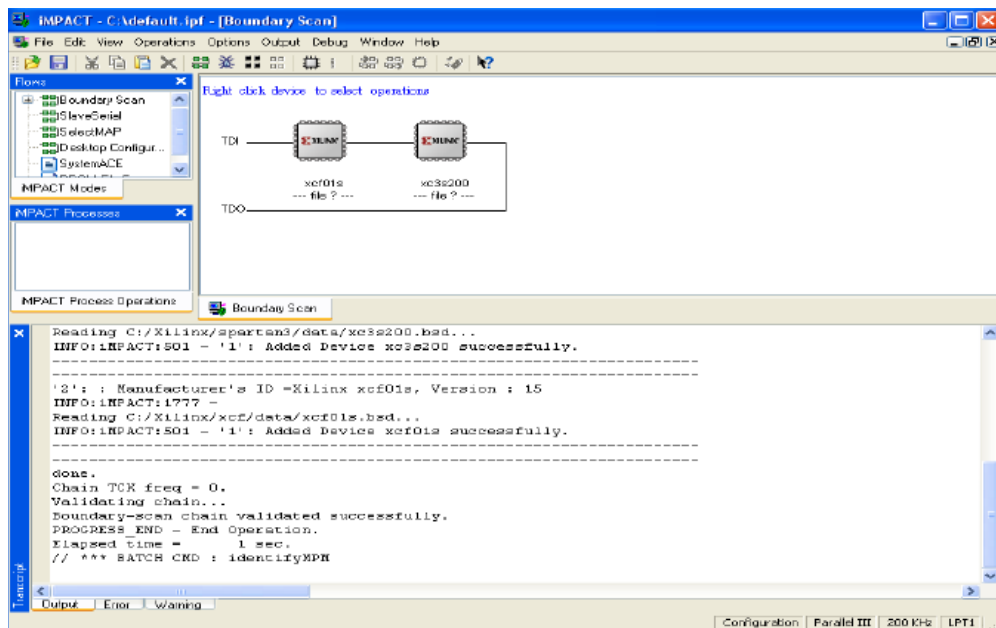


Fig. 4

Click twice on a chip model by right, which represent our chip and after Open File window will appear choose *Test_3.bit* . Below chip model should appear name of bit file.

Next click right mouse bottom on chip model, choose menu will appear from which select '*Program...*' . Appear window like in Figure 5 below:

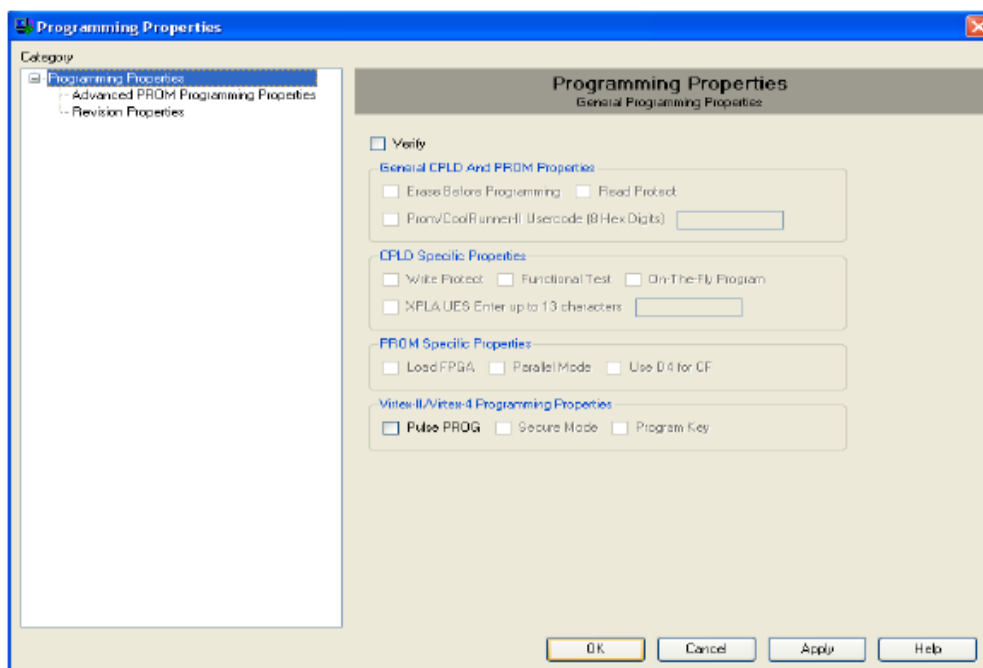


Fig. 5

Please don't change nothing and click 'OK' , which should load our project to chip. After correct programming will appear text '*Program Succeeded*' and we should seen working program on evaluation board.